Bone Age Prediction from Hand Radiographs

Pattern Recognition and Machine Learning Course Project

Syed Farhan Syed Sathik Basha (CS23B2039)
Bommireddy Raviteja Reddy (CS23B2011)
Mohamed Amjad (CS23B2013)

Department of Computer Science and Engineering

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Abstract

Bone age assessment is a critical clinical procedure for evaluating skeletal maturity in pediatric patients. Traditional manual methods are time-consuming and suffer from inter-observer variability. This project develops an automated deep learning system using ResNet50 CNN to predict bone age from hand X-rays. We implemented both regression (continuous age prediction) and classification (developmental stages) models on the RSNA Bone Age Dataset (12,611 images). The regression model achieved MAE = 9.55 months with $R^2 = 0.90$, while the classification model attained 88% accuracy. Gender-wise analysis revealed minimal bias (0.78 months difference). Grad-CAM visualizations confirm the model focuses on clinically relevant skeletal features.

1 Methodology

1.1 Dataset and Preprocessing

We used the **RSNA Bone Age Dataset** containing 12,611 hand X-ray images with bone age labels (in months) and biological sex metadata. Dataset split: 70% training (8,827), 15% validation (1,892), and 15% test (1,892) using stratified sampling.

Developmental Stage Labels:

- Infant: 0–24 months
- Child: 25–144 months
- Adolescent: 145–204 months
- Young Adult: >204 months

Preprocessing:

1. Images resized to 256×256 pixels (GPU memory constraints)
2. ImageNet normalization (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])
3. Data augmentation (training): random flips, rotations (±20°), color jitter
4. All images loaded into RAM (∼2.5GB) for faster training

1.2 Model Architecture

ResNet50 (pretrained on ImageNet) was used as backbone:

Regression Model:

- Modified head: Linear(2048 $\rightarrow$ 512) + ReLU + Dropout(0.3) $\rightarrow$ Linear(512 $\rightarrow$ 1)
- Loss: L1 Loss (MAE)
- Output: Predicted age in months

Classification Model:

- Modified head: Linear(2048 $\rightarrow$ 4)
- Loss: Cross-Entropy Loss
- Output: 4 developmental stage probabilities

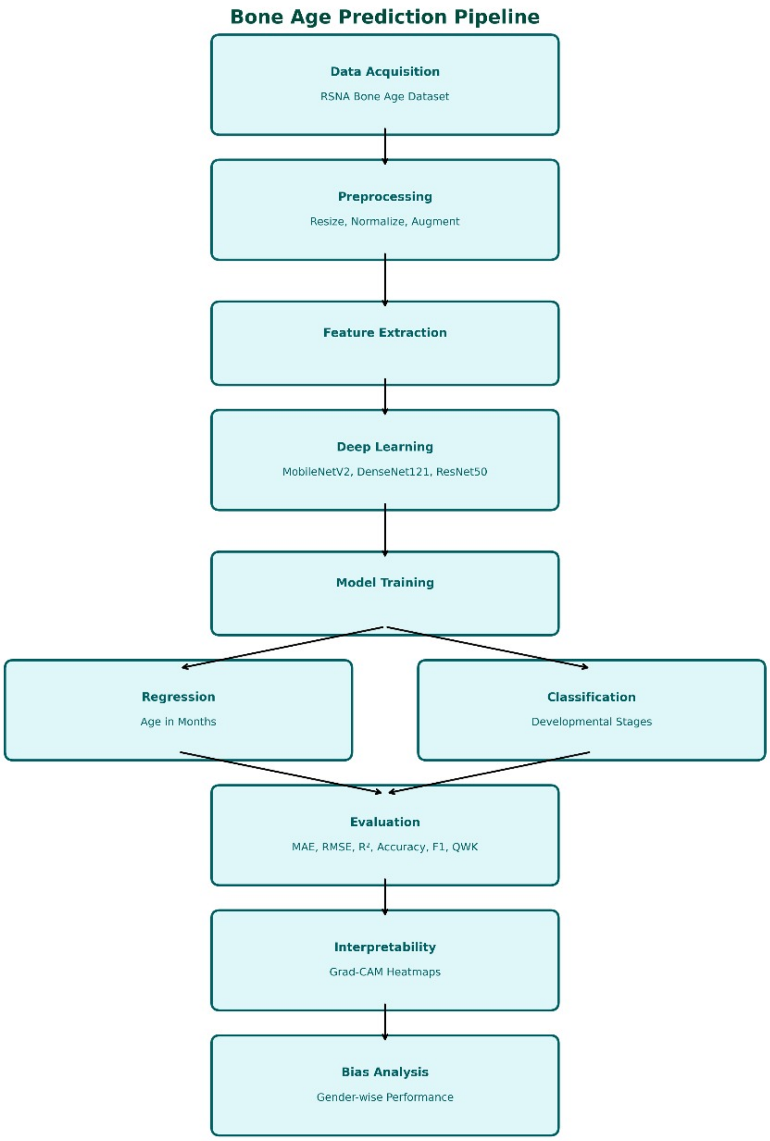

Figure 1: Pipeline for the models used

1.3 Training Configuration

- Optimizer: Adam (lr=1e-4)
- Batch size: 32, Epochs: 15
- LR Scheduler: ReduceLROnPlateau (factor=0.1, patience=3)
- Mixed Precision Training (AMP) enabled
- GPU: NVIDIA RTX 3050 Laptop
- Training time: ∼70 minutes

2 Results

2.1 Regression Performance

Metric	**Value**
MAE (months)	**9.55**
RMSE (months)	12.37
R^2 Score	0.90

Table 1: Regression Model Performance

MAE of 9.55 months indicates predictions within <1 year of true values, clinically acceptable and competitive with manual variability (12–18 months). High R^2 (0.90) explains 90% of variance.

2.2 Classification Performance

Metric	**Value**
Accuracy	**88%**
Quadratic Weighted Kappa	0.78
Precision (macro)	0.79
Recall (macro)	0.73
F1-Score (macro)	0.75

Table 2: Classification Performance

Confusion matrix showed highest accuracy for "Child" and "Adolescent" classes. Adjacent class confusion (e.g., Adolescent vs. Young Adult) is expected due to gradual developmental transitions.

2.3 Gender-Wise Analysis

Minimal performance disparity (<1 month) demonstrates fair generalization across demographics.

Gender	MAE (months)	Samples
Male	9.20	5,940
Female	9.98	6,671
Difference	**0.78**	–

Table 3: Gender-wise Performance

3 Comparison between different models

To justify selecting ResNet50 as our model, we compared three CNN backbones—ResNet50, DenseNet121, and MobileNetV2—trained for bone age prediction on hand radiographs. The evaluation was based on mean absolute error (MAE) for regression and classification accuracy for each model.

3.1 Plots and visualizations

Include the following figures as required: scatter plots of predicted vs. actual ages, Classification accuracies of different models.

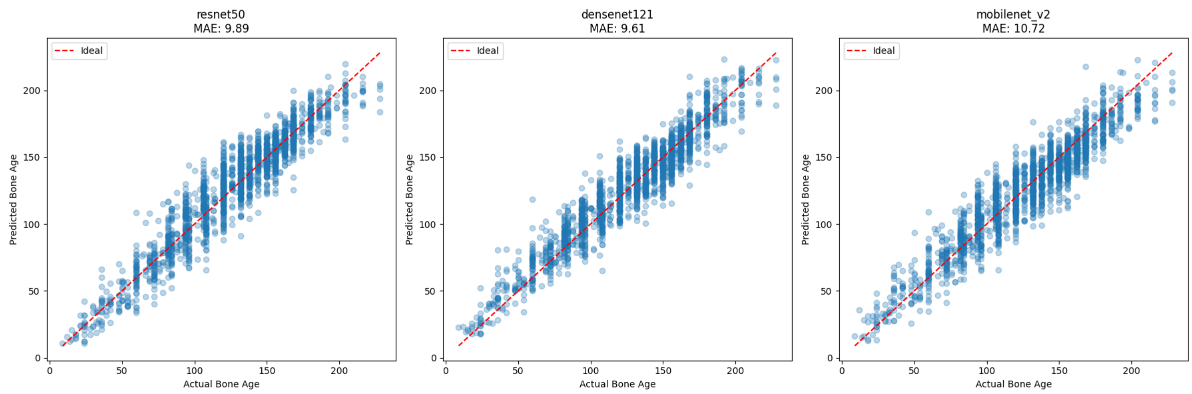

Figure 2: Predicted vs. true ages on the test set for each model.

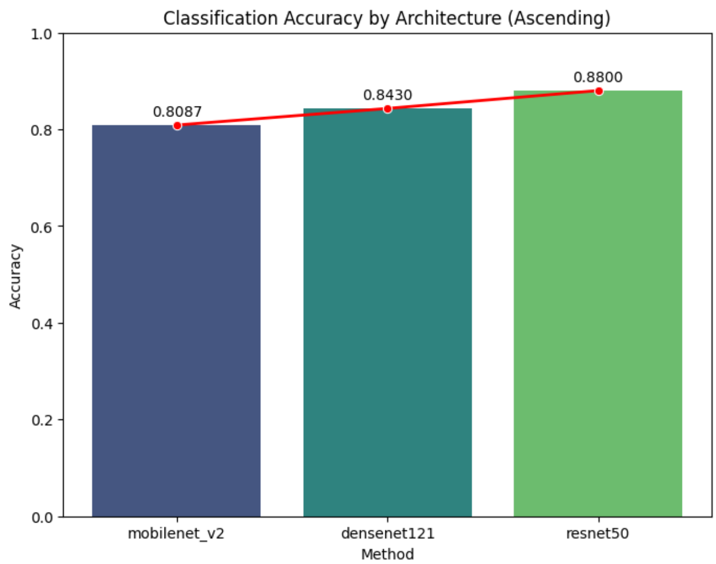

Figure 3: Accuracy on the test set for each model.

4 Discussion

4.1 Error Analysis

Key Findings:

1. MAE (9.55 months) within clinically acceptable range
2. Strong R^2 (0.90) indicates excellent model fit
3. Balanced gender performance (0.78 months difference)
4. High classification accuracy (88%) enables reliable staging

Error Patterns:

- **Age distribution:** Tighter predictions in 60–180 month range (Child/Adolescent) where training samples are abundant
- **Boundary effects:** Larger errors at extremes (very young infants, older adolescents) due to fewer examples and greater anatomical variability
- **Class confusion:** Adjacent developmental stages overlap in confusion matrix, reflecting continuous skeletal development

Insights from Model Behavior:

- Training curves showed consistent improvement without overfitting
- Validation MAE decreased from 19.31 (Epoch 1) to 9.55 (Epoch 10)
- Learning rate reduction at Epoch 10 led to fine-tuned convergence
- Classification model converged faster (stable by Epoch 8)

4.2 Model Interpretability (Grad-CAM Analysis)

Gradient-weighted Class Activation Mapping revealed:

- Model focuses on **wrist bones (carpals), metacarpals, phalanges** – same landmarks radiologists use in Greulich-Pyle assessments
- Attention patterns vary across stages: increased focus on epiphyseal plates in younger patients
- Validates learning of clinically meaningful features vs. spurious correlations

4.3 Limitations

1. **Resolution:** Downsampling to 256×256 may lose fine skeletal details
2. **Architecture:** Only ResNet50 explored; newer models (EfficientNet, ViT) might improve performance
3. **Error at extremes:** Need targeted augmentation for underrepresented age groups
4. **Generalization:** Cross-dataset validation needed

Conclusion

This project developed an automated bone age assessment system achieving MAE = 9.55 months and 88% classification accuracy with minimal gender bias. Grad-CAM confirms anatomically meaningful feature learning. The system offers clinical utility through speed, consistency, and scalability, though it should complement rather than replace expert judgment.

References

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